

Dynamic Games

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Sequential Games

- So far we've studied games where all players choose simultaneously.
- Now we study **dynamic games** (also called sequential games), where players move in a specific order.
- This requires a new representation: the **extensive form** or **game tree**.

Associated Reading

Game Theory, Sections 3.1–3.3

Why Sequential Order Matters

Compare:

- **Simultaneous:** You and a rival firm set prices at the same time. Neither knows the other's choice.
- **Sequential:** You set prices first. Your rival observes your choice, then sets theirs.

The sequential version gives the second mover an advantage. This is *strategically* very different.

The Extensive Form

A **game tree** represents a sequential game:

- **Nodes:** points where a player makes a decision.
- **Branches:** the possible actions at each node.
- **Terminal nodes:** the end of the game, where payoffs are assigned.
- **Player labels:** each non-terminal node is labeled with which player moves there.

A Simple Entry Deterrence Game

A potential **Entrant** decides whether to enter a market currently monopolized by an **Incumbent**.

If the Entrant enters, the Incumbent can either **Fight** (aggressive price war) or **Accommodate** (share the market peacefully).

Entrant

├ Stay Out $\rightarrow (0, 4)$ [Entrant gets 0; Incumbent keeps monopoly]

└ Enter

└ Incumbent

├ Fight $\rightarrow (-1, 1)$ [Price war: both lose]

└ Accommodate $\rightarrow (2, 2)$ [Both share market]

Payoffs: (Entrant, Incumbent)

A **strategy** for a player specifies what they will do at **every** decision node where they might move.

This includes nodes they might never actually reach. The strategy is a *complete contingent plan*.

For the Entrant: {Stay Out} or {Enter}.

For the Incumbent: {Fight} or {Accommodate} (only relevant if Entrant enters, but must be specified).

All Strategy Combinations

Entrant	Incumbent	Payoffs
Stay Out	Fight	(0, 4)
Stay Out	Accommodate	(0, 4)
Enter	Fight	(-1, 1)
Enter	Accommodate	(2, 2)

The Incumbent's strategy only matters when the Entrant actually enters.
But it's still a strategy choice that must be made (even if in advance).

Sometimes a player doesn't know where they are in the game tree. This is represented by an **information set**: a group of nodes among which the player can't distinguish.

A game has **perfect information** if every player always knows the full history of play when it is their turn.

Our examples so far are perfect information games. The Incumbent knows the Entrant has entered.

More Examples

The Ultimatum Game

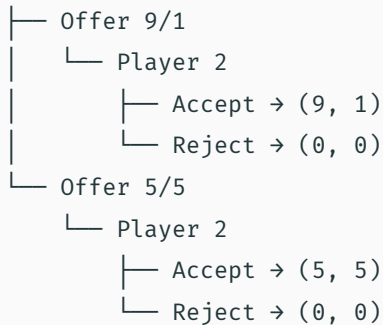
Player 1 proposes a split of \$10. Player 2 can Accept or Reject.

- If Accepted: both get the proposed split.
- If Rejected: both get \$0.

Assume P1 offers either 9/1 (keeping \$9) or 5/5 (equal split).

Ultimatum Tree

Player 1



Player 1's strategies: {Offer 9/1} or {Offer 5/5}.

Player 2's strategies: a *complete plan* for both nodes.

- Accept 9/1 and Accept 5/5
- Accept 9/1 and Reject 5/5
- Reject 9/1 and Accept 5/5
- Reject 9/1 and Reject 5/5

That's **4 strategies** for Player 2, even though only one node gets reached in any play of the game.

The Trust Game

Player 1 decides to **Trust** or **Not Trust** Player 2. If Player 1 trusts:

- The amount doubles (e.g., \$10 $\rightarrow$ \$20).
- Player 2 decides to **Honor** (split evenly) or **Exploit** (keep all).

Player 1

├ Not Trust $\rightarrow (5, 5)$

└ Trust

└ Player 2

├ Honor $\rightarrow (10, 10)$

└ Exploit $\rightarrow (0, 20)$

Backward Induction

How should we analyze these sequential games? The method is **backward induction**: start at the final decision nodes and work towards the root.

At each step, assume the player at that node will choose the action that maximises their own payoff. Replace the node with the resulting payoff, and continue.

Backward Induction in the Entry Game

Entrant

├ Stay Out $\rightarrow (0, 4)$

└ Enter

└ Incumbent

├ Fight $\rightarrow (-1, 1)$

└ Accommodate $\rightarrow (2, 2)$

Step 1: At the Incumbent's node, Fight gives 1 and Accommodate gives 2.
Rational choice: **Accommodate**.

Step 2: Replace the Incumbent's node with (2, 2). The Entrant chooses between Stay Out (0) and Enter (2). Rational choice: **Enter**.

Outcome: Enter $\rightarrow$ Accommodate, with payoffs (2, 2).

Backward Induction in the Trust Game

Player 1

├ Not Trust $\rightarrow (5, 5)$

└ Trust

└ Player 2

├ Honor $\rightarrow (10, 10)$

└ Exploit $\rightarrow (0, 20)$

Step 1: Player 2 gets 10 from Honor vs. 20 from Exploit. A rational Player 2 chooses **Exploit**.

Step 2: Player 1 anticipates this. Trusting leads to 0; Not Trusting gives 5. Player 1 chooses **Not Trust**.

Outcome: (5, 5). Yet (Trust, Honor) would give (10, 10), better for both. This tension between individual rationality and mutual benefit echoes the Prisoner's Dilemma.

Worked Example

A Hiring Game

A firm (Player 1) interviews a candidate and decides to **Offer** or **Pass**. If the firm offers, the candidate (Player 2) can **Accept** or **Negotiate**. If the candidate negotiates, the firm can **Raise** the offer or **Withdraw** it.

Firm

├─ Pass $\rightarrow (0, 2)$

└─ Offer

└─ Candidate

└─ Accept $\rightarrow (3, 3)$

└─ Negotiate

└─ Firm

└─ Raise $\rightarrow (1, 5)$

└─ Withdraw $\rightarrow (-1, 0)$

Payoffs: (Firm, Candidate)

Part 1: Strategies

How many strategies does the Firm have?

The Firm moves at two nodes: the root, and after a negotiation. So the Firm's strategies are complete plans for both nodes.

- (Pass; Raise)
- (Pass; Withdraw)
- (Offer; Raise)
- (Offer; Withdraw)

That's **4 strategies** for the Firm. Notice that (Pass; Raise) and (Pass; Withdraw) are different strategies, even though the second part never comes up if the Firm passes. A strategy is a complete plan that specifies what to do at every decision node, including nodes that the strategy itself makes unreachable.

The Candidate has two nodes but only one is a decision node, so 2

Part 2: Backward Induction

Step 1: At the Firm's second node, Raise gives 1 and Withdraw gives -1. The Firm chooses **Raise**.

Step 2: Replace that node with (1, 5). The Candidate now faces Accept (payoff 3) vs. Negotiate (payoff 5). The Candidate chooses **Negotiate**.

Step 3: Replace the Candidate's node with (1, 5). The Firm faces Pass (payoff 0) vs. Offer (payoff 1). The Firm chooses **Offer**.

Outcome: Offer → Negotiate → Raise, with payoffs (1, 5).

What Can We Learn?

The Candidate gets their best outcome by negotiating, because they can anticipate the Firm will raise rather than withdraw.

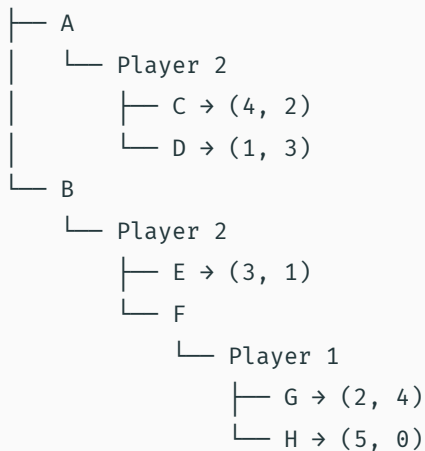
The Firm would prefer $(3, 3)$ from an immediate acceptance, but backward induction tells us the Candidate won't accept the initial offer.

This is a common pattern: the player who moves later can sometimes extract a better deal, because the earlier player's threat to withdraw is costly to carry out.

Another Worked Example

Consider the following game:

Player 1

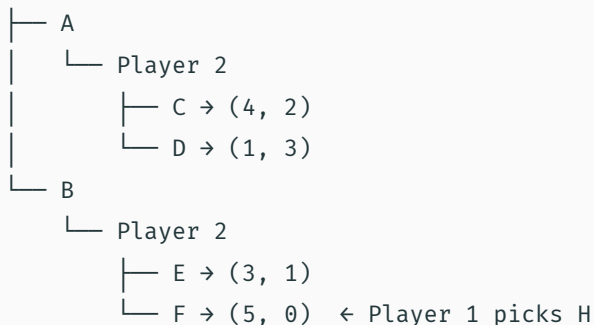


Payoffs: (Player 1, Player 2)

Part (a): Step 1

Start at the deepest node: Player 1's second decision (after B, F). G gives Player 1 a payoff of 2, H gives 5. Player 1 chooses **H**. Replace that node with (5, 0).

Player 1



Part (a): Steps 2 and 3

Now solve both of Player 2's nodes. After A: C gives Player 2 a payoff of 2, D gives 3. Player 2 chooses **D**. After B: E gives 1, F gives 0. Player 2 chooses **E**.

Player 1

├ A $\rightarrow (1, 3)$ $\leftarrow$ Player 2 picks D
└ B $\rightarrow (3, 1)$ $\leftarrow$ Player 2 picks E

Part (a): Step 4

Player 1 faces $A \rightarrow (1, 3)$ vs $B \rightarrow (3, 1)$. Player 1 gets 1 from A and 3 from B.
Player 1 chooses **B**.

Outcome: $B \rightarrow E$, with payoffs **(3, 1)**.

Part (b): The Outcome

The backward-induction outcome is $B \rightarrow E$, with payoffs $(3, 1)$.

Player 2 chooses E because the alternative (F) leads to Player 1 choosing H, which gives Player 2 a payoff of 0. Player 2 would prefer D (payoff 3) if Player 1 played A, but Player 1 won't play A (payoff 1 vs 3).

Part (c): Strategy Profiles

Player 1 moves at two nodes: the root, and after (B, F). So Player 1's strategies have two components.

The backward-induction strategy for Player 1 is **(B; H)**: play B at the root, and play H if the game ever reaches the second node.

Player 2 moves at two nodes: after A, and after B. The backward-induction strategy for Player 2 is **(D; E)**: play D after A, and play E after B.

The full strategy profile is **(B; H), (D; E)**. Note that Player 1's second node is never reached on the path of play, but the strategy still specifies H there. And Player 2's choice of D after A is never reached either. Both of these off-path choices matter for the analysis, because they are what make the equilibrium subgame perfect.

- Backward induction gives us a solution concept: **subgame perfect equilibrium**.
- We'll see how it rules out Nash equilibria that rely on empty threats.
- And we'll look at **win-lose games** where backward induction has clean results.